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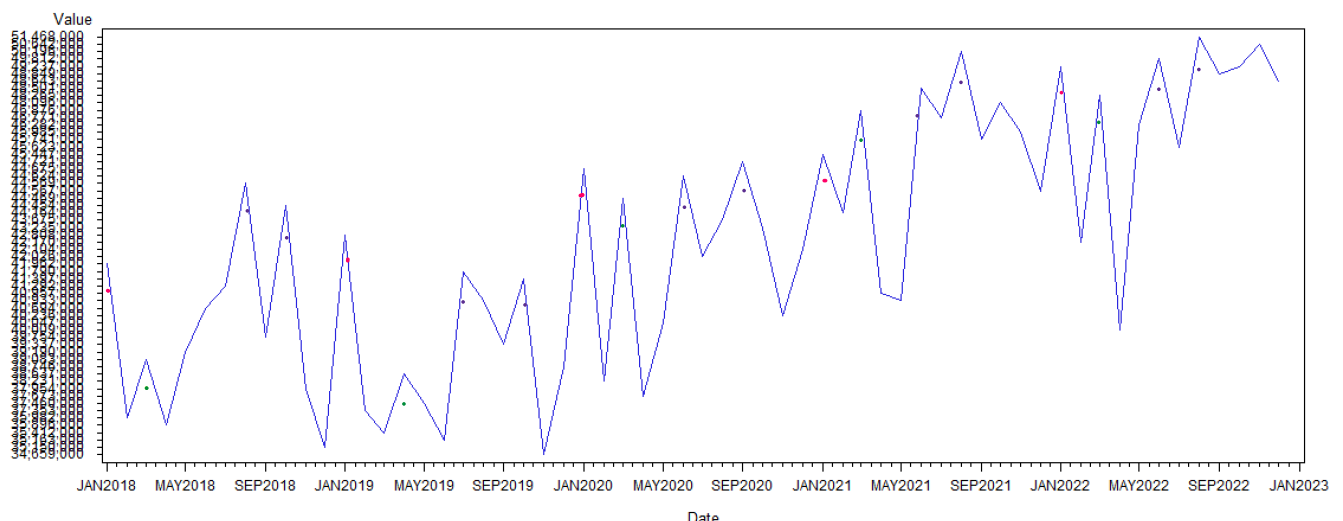
3 August 2025

Pennsylvanian Chicken Production Data over Five Years Before and After COVID-19 Pandemic

Introduction

I chose to observe the total chicken production spanning from 2018 to 2022 in Pennsylvania, a total of five years' worth of data. The objective was to see how chicken production was affected before, during, and after the initial waves of the quarantines, slowed production, and economic turndowns of the COVID-19 pandemic. The data was obtained via the USDA Agricultural database. Three different types of analyses were performed on the selected data, a seasonal decomposition, exponential smoothing, and a Box and Jenkins model. In addition, forecasts and prediction intervals were provided for three time periods into the future. Since the time intervals provided were monthly basis, this will be the first three months into the future from this range, 2023.

Figure 1: Monthly Total Production of Chickens across Pennsylvania from January 2018 to December 2022. Seasonal spikes marked in red (January), green (spring), and purple (, before and after September pre-2020, May to September post-2020).



Seasonal Decomposition Modeling (A)

Based on the appearance of Figure 1, seasonal decomposition multiplicative modeling was more appropriate. This is due to observing the data is roughly seasonal. There are regular seasonal spikes every January, spring months (usually March sometimes April), two spikes in the summer months, and dips in autumn months right before the following January. In addition, this seasonal trend is also increasing linearly, as visible each January spike is higher than the next, which makes sense since chicken demand would increase with a growing population or population demands.

Conducted Seasonal Decomposition Modeling by way of proc timeseries and proc reg within SAS:

Figure A1: SAS output from the regression of deseasonalized values timeseries. Intercept value and slope value are highlighted in red descending in column Parameter Estimate, respectively.

Model: MODEL1

Dependent Variable: SA Seasonally Adjusted Series

Number of Observations Read	60
Number of Observations Used	60

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	6.650129E14	6.650129E14	135.58	<.0001
Error	58	2.8449E14	4.905001E12		
Corrected Total	59	9.495029E14			

Root MSE	2214724	R-Square	0.7004
Dependent Mean	42743207	Adj R-Sq	0.6952
Coeff Var	5.18146		

Parameter Estimates						
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	Intercept	1	36879950	579063	63.69	<.0001
t		1	192238	16510	11.64	<.0001

Figure A2: SAS output from the timeseries procedure. January, February, and March seasonal estimates are highlighted in red descending in column SC, respectively.

Obs	_NAME_	_MODE_	Date	_SEASON_	ORIGINAL	TCC	SC	IC	SA	t
1	Chickens	MULT	JAN2018	1	41962000	.	1.07300	.	39107070.17	1
2	Chickens	MULT	FEB2018	2	35982000	.	0.95049	.	37856318.41	2
3	Chickens	MULT	MAR2018	3	39083000	.	1.02039	.	38302170.54	3
4	Chickens	MULT	APR2018	4	35896000	.	0.91939	.	39043167.14	4
5	Chickens	MULT	MAY2018	5	39190000	.	0.95592	.	40997119.89	5
6	Chickens	MULT	JUN2018	6	40694000	.	1.02256	.	39796224.58	6
7	Chickens	MULT	JUL2018	7	41292000	39700250.00	1.03360	1.00629	39949858.88	7
8	Chickens	MULT	AUG2018	8	44569000	39792625.00	1.07355	1.04329	41515376.68	8
9	Chickens	MULT	SEP2018	9	39754000	39696791.67	1.01089	0.99065	39325737.46	9
10	Chickens	MULT	OCT2018	10	44454000	39658041.67	1.05427	1.06323	42165524.69	10
11	Chickens	MULT	NOV2018	11	37954000	39700166.67	0.93931	1.01778	40406111.63	11
12	Chickens	MULT	DEC2018	12	35150000	39397625.00	0.94662	0.94249	37131997.68	12
13	Chickens	MULT	JAN2019	1	42808000	39187916.67	1.07300	1.01806	39895511.65	13

Figure A3: SAS output from the regression of Deseasonalized values for prediction intervals (blue). January, February, and March are Observations 61, 62, and 63.

60	51385825	48414226	564765	43839104	52989348	2971599
61	.	48606464	579063	44024186	53188742	.
62	.	48798702	593476	44209041	53388363	.
63	.	48990940	607995	44393671	53588210	.

Forecast and Prediction Intervals 3 Time Periods into the Future (Manually):

Using the highlighted values in Figures A1, A2, and A3, the forecasted reseasonalized values and prediction intervals were determined as follows.

Forecasted value, $yt=dt*snt$

Prediction interval, $yt \pm MOE$. Where $MOE = (Upper\ Bound - Lower\ bound)/2$

Month Period (2023)	Reseasonalized Forecast	Margin of Error	Prediction Interval
January	52154735	4582278	[47572457, 56737013]
February	46382678	4589661	[41793017, 50972339]
March	49989865	4597270	[45392596, 54587135]

The results of this modeling shows that multiplicative was the best approach compared to additive, as far as seasonal decomposition goes. Figure A1 output verifies that the regression itself is statistically significant due to the model's low p-value well below the typical 0.05 critical value, as well as for the intercept and the slope parameter estimates.

Exponential Smoothing Modeling (ESM) (B)

Based on Figure 1 appearance, Holt-Winters multiplicative model might be the best suited exponential smoothing. This is due to observing the data is roughly seasonal; there is a trough dip occurring around autumn, then sharply climbing in January of the next year therefore a seasonal variation. This leaves likely either of the two Holt-Winters models or the damped model.

Furthermore, between each season the height of the peak from the baseline, (amplitude) is increasing visibly. Since this is an increasing seasonal variation, this leaves Holt-Winters Multiplicative ESM as the best option, eliminating the H-W Additive ESM. Holt-Winters Multiplicative is almost always chosen when there is increasing or decreasing multiplicative seasonality (over damped trend model).

Conducted Holt-Winters multiplicative ESM by way of proc ESM within SAS:

Figure B1: SAS output from the proc ESM/Model Viewer TSFS.

Parameter Estimates				
CHICKENS				
Winters Method -- Multiplicative				
Model Parameter	Estimate	Std. Error	T	Prob> T
LEVEL Smoothing Weight	0.19115	0.0566	3.3788	0.0013
TREND Smoothing Weight	0.00100	0.0209	0.0478	0.9621
SEASONAL Smoothing Weight	0.33012	0.1050	3.1434	0.0027
Residual Variance (sigma squared)	5.62805E12	.	.	.
Smoothed Level	49722858	.	.	.
Smoothed Trend	188765	.	.	.
Smoothed Seasonal Factor 1	1.06669	.	.	.
Smoothed Seasonal Factor 2	0.94130	.	.	.
Smoothed Seasonal Factor 3	1.01885	.	.	.
Smoothed Seasonal Factor 4	0.90446	.	.	.
Smoothed Seasonal Factor 5	0.96119	.	.	.
Smoothed Seasonal Factor 6	1.02992	.	.	.
Smoothed Seasonal Factor 7	1.00781	.	.	.
Smoothed Seasonal Factor 8	1.07473	.	.	.
Smoothed Seasonal Factor 9	1.01193	.	.	.
Smoothed Seasonal Factor 10	1.04192	.	.	.
Smoothed Seasonal Factor 11	0.97009	.	.	.
Smoothed Seasonal Factor 12	0.95494	.	.	.

- The smoothing constants are the parameter estimates boxed in red:
 - level smoothing constant $\alpha = 0.19115$,
 - growth rate smoothing constant $\gamma = 0.00100$, and
 - seasonal factor smoothing constant $\delta = 0.33012$.
- Observing the right-most column of Figure 2 boxed in red, level and seasonal factor smoothing constants have very low p-values 0.0013 and 0.0027, considerably less than the typically referenced critical value $\alpha = 0.05$. However, growth rate smoothing constant has a p-value well above 0.05, suggesting the trend is not statistically significant under exponential smoothing. Since there is still a seasonal component to the data and statistically verified by the parameter's p-value, this suggest Holt-Winters Multiplicative model was still appropriate as far as exponential smoothing goes.
- The final estimates for the level at most recent time point is $l_T = 49,722,858$ chickens, boxed in blue. The final estimate for the slope at most recent time point, is boxed in yellow, $b_T = 188,765$ chickens/month. The final estimates for the seasonality at most recent time point of each month 1 through 12, sn_T , are boxed in green.

Forecast and Prediction Intervals 3 Time Periods into the Future (SAS):

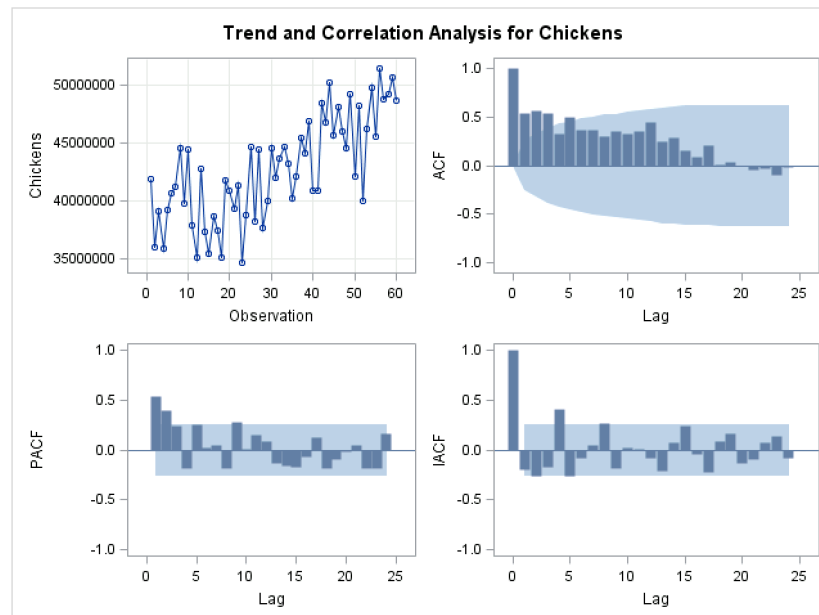
Figure B2: SAS output of proc ESM for Holt-Winters Multiplicative model for forecasted values and prediction intervals at 95% confidence in number of chickens.

Forecasts for Variable Chickens					
Obs	Time	Forecasts	Standard Error	95% Confidence Limits	
61	JAN2023	53240126.97	2372350.3	48590405.86	57889848.08
62	FEB2023	47159688.40	2405931.2	42444150.00	51875226.81
63	MAR2023	51236893.05	2461260.9	46412910.43	56060875.68

Box and Jenkins Modeling (C)

Based on the findings from Figure 1, a mixed ARIMA model will be employed. First, a log transformation of the data was applied, due to the increasing trend of the data and thus non-stationarity. Then, by use of proc ARIMA, it was determined by the ACF and PACF plots that a mixed ARIMA model would be best, as depicted in Figure C1. As visible, a clear upward trend still suggested differencing was needed ($d=1$, $D=1$, and $S=12$), and the ACF plot shows seasonal lags and non-stationarity, strengthening the mixed ARIMA parameters. PACF plot also shows a significant spike at lag 1, which suggested ARIMA (1) or (2), thus ARIMA(3,1,0)(0,1,0) was chosen and applied via TSFS.

Figure C1: SAS Output of ARIMA procedure of log-transformed data.



Then, the model parameter estimates were determined by TSFS applying ARIMA mixed model and including the forecasting for three time periods in the future (January-March 2023). The ARIMA mixed model was strengthened based on the plots from TSFS as depicted in Figure C2. ACF indicates no significant autocorrelation remains, PACF indicates the residuals behave like white noise (all contained within ± 0.5) and IACF indicates a stable model. Figure C3 indicates the AR lags 1 and 2 were statistically significant, however lag 3 and intercept parameter estimates were not.

Figure C2: SAS Output from TSFS For Mixed ARIMA of Log-transformed data.

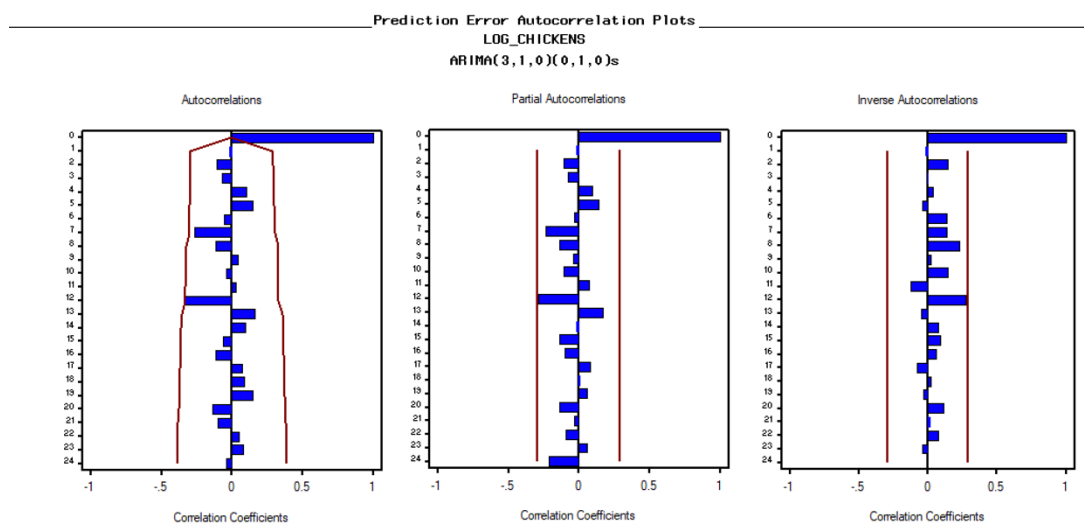


Figure C3: Parameter Estimates for Mixed ARIMA.

Parameter Estimates				
LOG_CHICKENS				
ARIMA(3,1,0)(0,1,0)s				
Model Parameter	Estimate	Std. Error	T	Prob> T
Intercept	0.00173	0.0042	0.4173	0.6785
Autoregressive, Lag 1	-0.90439	0.1514	-5.9732	<.0001
Autoregressive, Lag 2	-0.58560	0.1841	-3.1806	0.0027
Autoregressive, Lag 3	0.04084	0.1513	0.2700	0.7885
Model Variance (sigma squared)	0.00484	.	.	.

Forecast and Prediction Intervals 3 Time Periods into the Future (SAS):

Data output from forecast of TSFS as depicted in Figure C4 then had to be exponentially-transformed due to being log-transformed values, as depicted in Figure C5. Performed in SAS by code.

Figure C4: SAS TSFS Output of Forecast Predicted Values and Prediction Intervals at 95% Confidence Interval for 2023.

Forecast Data Set						
LOG_CHICKENS						
ARIMA(3,1,0)(0,1,0)s						
DATE	ACTUAL	PREDICT	U95	L95	ERROR	NERROR
JAN2023	.	17.7675	17.9039	17.6312	.	.
FEB2023	.	17.6541	17.7911	17.5171	.	.
MAR2023	.	17.7743	17.9184	17.6302	.	.

Figure C5: SAS Output of Exponentially-Transformed Predicted Values and Prediction Intervals at 95% Confidence Interval for 2023.

Obs	predict_exp	L95_exp	U95_exp
61	52040067.27	59642752.69	45406499.19
62	46461305.28	53282098.50	40513661.23
63	52391913.12	60512293.68	45361238.08

Conclusion and Argument

Based on the three analyses performed, the best model deployed was multiplicative seasonal decomposition, because it was the only modeling system where all its parameters were statistically significant, as well as the model's p-value. In addition, it was the only one that involved the least manipulation of data, such as the log transformation that took place in the mixed ARIMA of Box and Jenkins modeling. Also, the mixed ARIMA show a statistically insignificant parameter estimate for the intercept, which would not make sense due to the increasing trend observed in Figure 1. While the Exponential Smoothing H-W did not involve data transformation or loss of information, however its trend parameter estimate was not statistically significant which doesn't make sense due to chicken production clearly increasing over time, as would be expected in a general population. Thus, Multiplicative Seasonal Decomposition in my opinion was the best fit.